

KIGALI INTL, Rwanda

WMO: 643870

Lat: 1.969S

Lon: 30.139E

Elev: 4859

StdP: 12.29

Time Zone: 2.00 (E02)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	58.7	59.3	44.3	51.5	76.6	46.4	55.7	75.0	19.1	71.0	15.3	72.5	2.7	320	0.266

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 19.4	(c) 86.3	(d) 65.5	(e) 84.9	(f) 65.4	(g) 84.0	(h) 65.3	(i) 70.5	(j) 78.7	(k) 69.9	(l) 78.3	(m) 69.3	(n) 77.7	(o) 7.8	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.3	124.8	72.4	67.9	123.2	72.1	66.6	117.7	71.2	37.8	78.6	37.1	78.6	36.6	78.0	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 14.8	(b) 12.2	(c) 10.9	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) 14.8	(b) 12.2	(c) 10.9	(d) WB	(e) 50.7	(f) 72.3	(g) 1.7	(h) 1.3	(i) 49.5	(j) 73.2	(k) 48.5	(l) 74.0	(m) 47.6	(n) 74.7	(o) 46.4	(p) 75.6	(q) 75.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	71.2	72.1	72.1	71.2	70.4	70.5	70.6	71.0	72.4	72.1	71.6	70.2	70.7
	DBStd	2.28	2.35	2.64	2.60	2.13	1.63	1.57	1.96	2.10	2.16	2.38	1.98	2.08
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	7750	685	619	657	612	636	617	651	695	662	669	606	641
	CDD65	2278	220	199	193	162	171	167	186	231	213	204	156	176
	CDH74	11892	1316	1227	1007	565	586	822	1187	1583	1237	1036	563	763
	CDH80	1984	309	319	190	38	27	49	145	316	243	190	54	104
(14) Wind	WSAvg	4.4	4.2	4.1	4.1	4.3	4.1	4.3	4.5	5.1	5.1	4.8	4.2	4.1
(15) Precipitation	PrecAvg	47.0	3.4	3.3	5.8	6.1	4.7	1.4	1.1	2.4	3.7	4.9	5.7	4.4
	PrecMax	62.7	7.0	9.4	10.0	9.5	8.1	3.0	3.7	4.1	8.0	10.4	8.9	9.4
	PrecMin	39.0	0.9	0.8	2.8	2.8	2.3	0.1	0.1	0.6	1.5	2.3	2.9	1.4
	PrecStd	5.4	1.3	1.6	1.5	1.4	1.4	0.8	0.7	0.9	1.3	1.5	1.5	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.9	89.4	86.3	82.7	82.5	82.8	84.7	87.4	86.3	86.2	83.8	84.6
		MCWB	64.8	65.0	67.6	68.7	67.8	65.6	63.6	63.5	65.0	65.3	68.4	67.0
	2%	DB	85.9	86.2	84.2	80.9	80.6	81.0	82.8	84.6	84.4	84.2	81.3	82.6
		MCWB	66.0	66.0	67.6	68.2	67.2	64.7	62.5	62.7	64.8	65.7	67.9	67.1
	5%	DB	83.9	84.0	82.3	79.1	78.9	80.4	81.8	83.0	82.7	82.3	79.8	80.7
		MCWB	65.9	66.0	67.7	68.0	67.1	64.5	62.3	62.6	64.4	65.7	67.7	67.0
	10%	DB	81.0	81.0	80.3	77.3	77.3	78.8	80.3	82.0	80.9	80.3	77.3	78.8
		MCWB	66.6	66.8	67.6	67.5	66.6	64.0	62.1	62.6	64.5	65.8	67.3	66.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.1	70.9	70.9	70.8	70.5	69.2	66.6	67.6	69.2	70.2	70.7	70.9
		MCDB	79.6	80.3	79.9	78.4	77.0	77.2	77.2	78.5	80.0	78.8	78.7	78.2
	2%	WB	69.9	69.6	69.8	69.8	69.3	67.9	65.0	66.5	68.1	69.2	69.6	69.7
		MCDB	78.4	79.2	78.9	77.8	76.4	76.2	76.7	77.3	78.4	77.7	77.6	77.7
	5%	WB	69.0	68.8	69.0	69.0	68.5	67.0	64.0	65.4	67.2	68.2	68.9	68.9
		MCDB	77.8	78.3	78.1	76.6	75.4	74.9	76.7	75.9	76.9	76.7	76.6	76.8
	10%	WB	68.0	67.9	68.3	68.2	67.7	65.9	63.0	64.5	66.3	67.3	68.0	68.0
		MCDB	77.2	77.2	77.2	75.4	74.2	73.8	76.6	74.7	75.4	75.4	75.5	75.9
(35) Mean Daily Temperature Range	5% DB	MDBR	19.4	19.4	18.0	15.6	15.6	17.6	19.5	19.4	19.1	18.3	16.9	17.3
		MCDBR	23.1	23.7	21.7	18.0	17.6	19.9	21.7	22.0	21.7	21.1	19.6	20.5
		MCWBR	7.9	7.9	8.1	7.3	6.5	6.5	7.1	6.8	6.7	7.2	7.8	7.5
	5% WB	MCDBR	19.4	19.5	19.1	16.7	15.8	17.1	19.5	18.8	19.2	18.4	17.8	18.0
		MCWBR	7.8	7.6	7.8	7.3	6.6	6.5	6.6	5.9	6.8	7.6	7.9	7.5
(40) Clear-Sky Solar Irradiance	taub		0.459	0.446	0.401	0.354	0.370	0.408	0.457	0.456	0.411	0.385	0.368	0.408
	taud		2.167	2.215	2.384	2.544	2.464	2.310	2.127	2.145	2.297	2.374	2.453	2.329
	Ebn at Noon		278	283	292	300	287	271	259	266	285	297	304	292
	Edh at Noon		50	48	40	33	35	40	48	49	43	41	37	42
(44) All-Sky Solar Radiation	RadAvg		1543	1601	1539	1472	1480	1571	1663	1600	1597	1529	1427	1461
	RadStd		121	88	86	92	64	105	101	111	88	90	73	101

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page